`nsm`: an explainable, test-backed consistency engine for Standard-Model and beyond-Standard-Model gauge theories

Evan Morritse

June 25, 2026

Abstract

We describe `nsm`, a symbolic-inference engine that treats the internal consistency of a gauge theory as a constraint-satisfaction problem over an SMT solver and elementary representation theory. From a representation skeleton together with the gauge invariance of the Yukawa couplings and one normalization convention, `nsm` derives which quantum numbers are *forced* (and reports those that are non-unique, solver-undetermined, or inconsistent); classifies which processes are allowed, forbidden, or suppressed, with a traceable reason; evaluates hypothesized new mediators against the *full* two-$U(1)$ anomaly set, including the mixed coefficients that hand checks frequently omit; and, given an inconsistent gauge extension, classifies the smallest *tier* of new matter that repairs it (a tiered menu from sterile singlets to SM-vector-like exotics), with a solver-backed record (an `unsat` ladder) that records whether smaller repairs are excluded within that menu. Every output is a structural statement, not a numerical observable. We validate the engine by reproducing known results: the Standard-Model hypercharges follow from anomaly cancellation and Yukawa gauge invariance, and for gauged $B-L$ the minimal one-field sterile completion is one right-handed neutrino per generation: rediscovered, not assumed. We emphasize what the engine does *not* do (no cross sections, no kinematics, no claim of existence) and describe the adversarial, test-backed methodology used to check each layer. `nsm` is a methods/tools contribution: an explainable "theory debugger," not a calculator.

1 Introduction

Most computational tools for particle theory are *calculators* (amplitudes, cross sections, decay rates) or *databases* (particle properties, model files). Comparatively little tooling answers the prior, structural questions a model builder actually asks first: *Given this gauge structure, what is forced? What is merely conventional? Is this proposed extension even consistent, and if not, what is the smallest thing that fixes it?*

`nsm`[1] is an attempt to make those questions mechanical and explainable. It encodes gauge-theoretic consistency (anomaly cancellation, gauge invariance of couplings, conservation laws, locality, unitarity-cut structure) as constraints over the SMT solver Z3 [1] together with explicit representation-theory data. The design goal is that every result be a *traceable structural statement* ("this charge is forced because dropping it leaves these anomaly coefficients nonzero"), rather than a floating-point number whose provenance is opaque.

Contributions.

[1] `nsm` is the package name, not an acronym; the project began as a "non-standard model" pun and is now a general consistency engine for Standard-Model and BSM gauge theories.

1. A single explanation pipeline spanning layered algebraic and graph-search engines: *detect* inconsistency, *explain* the obstruction, *search* a minimal repair, *re-check*, and *expose* structural process implications.

2. A two-$U(1)$ anomaly engine computing the non-abelian, gravitational, cubic, and mixed $[Y^2X]$, $[YX^2]$ coefficients.

3. A minimal-repair search over a tiered menu (sterile singlets and SM-vector-like, $U(1)_X$-chiral exotic pairs) that classifies the smallest tier of new matter an anomalous extension requires, with a solver-backed record reporting minimality within that menu: a consistency checker that doubles as a modest designer.

4. Validation by reproduction of known results and by adversarial tests in which each layer is independently re-derived.

Non-contributions. `nsm` computes no physical observables and makes no claim of discovery. It does not derive masses, mixing angles, coupling values, or the number of generations (these remain empirical inputs). Its results are structural and quantum-number-level; see Section 11. Throughout we use the convention $Q = T_3 + Y/2$, so the quark doublet carries $Y = 1/3$.

2 Design

`nsm` is organized as layers over one explanation pipeline (Table 1). Lower layers expose primitives (anomaly coefficients, a tree-level reachability search) that upper layers reuse, so that, e.g., a candidate gauge boson's consistency check is *literally* the kernel's anomaly-cancellation routine, and the process-mediation test is a special case of factorization-channel enumeration.

Layer	Establishes
kernel	hypercharges/charges *forced* by anomaly cancellation + Yukawa gauge invariance; reports unique / non-unique / solver-undetermined / inconsistent
dynamics	layered process verdict from conservation laws + a tree-level vertex search
extensions	field type of a candidate mediator (implemented coupling templates); consistency (gauge candidates reuse the anomaly engine); process exposure
amplitudes	leading soft factors and their Ward identities
factorization	internal lines a process factorizes through (locality)
unitarity	intermediate states a process can pass through (optical-theorem cuts), ordered by mass threshold
repair	smallest addition that makes an inconsistent gauge extension consistent

Table 1: The layers of `nsm`. Each is independently re-derived and tested.

3 Related work

`nsm` is closest in spirit to model-building infrastructure, but it is aimed at a different phase of the workflow. FeynRules [2] and SARAH [3] take a specified model (fields, symmetries, parameters, and often a Lagrangian) and derive model data such as vertices, mass matrices, renormalization-group equations, and interfaces to simulation tools. PyR@TE focuses on automatic renormalization-group equations for general gauge theories [4]. SModelS decomposes BSM collider signatures into simplified-model topologies and confronts them with LHC limits [5]. These tools are deeper calculators and phenomenology interfaces than `nsm` is.

The anomaly-equation literature addresses closely related mathematics, including general solutions of chiral $U(1)$ anomaly equations [6]. `nsm` does not compete with those classifications. Its contribution is instead an explainable inverse workflow: diagnose why a proposed extension is inconsistent, search a declared repair menu, record why smaller repairs fail, re-check the result, and connect the repaired mediator to structural process exposure. This is a deliberately narrower, more inspectable layer that can sit before a full model is handed to a calculator such as FeynRules or SARAH.

4 Forced quantum numbers

Electric charge is not a primitive input: it follows from weak isospin and hypercharge through $Q = T_3 + Y/2$. The kernel leaves all hypercharges *unknown* and imposes consistency as constraints:

- anomaly cancellation: the non-abelian, mixed, gravitational, and cubic coefficients must vanish, summed over Weyl fermions with a chirality sign $s = \pm 1$;
- gauge invariance of each Yukawa coupling (so fermions can acquire mass);
- one normalization ($Y_H = 1$).

Z3 then returns the hypercharges. For one Standard-Model generation the solution is unique:

$$Y_{Q_L} = \tfrac{1}{3}, \quad Y_{u_R} = \tfrac{4}{3}, \quad Y_{d_R} = -\tfrac{2}{3}, \quad Y_{L_L} = -1, \quad Y_{e_R} = -2, \tag{1}$$

whence $Q_u = +\frac{2}{3}$, $Q_d = -\frac{1}{3}$, $Q_\nu = 0$, $Q_e = -1$.

The value of the constraint formulation is that it equally reports what is *not* forced. Dropping the Yukawa couplings leaves an exact two-fold $u_R \leftrightarrow d_R$ hypercharge ambiguity; adding a free right-handed neutrino opens a one-parameter flat direction (gauged $B-L$); a lone colored Weyl fermion is rejected because $[SU(3)^3] \neq 0$; an odd number of weak doublets is rejected by the Witten $SU(2)$ global anomaly [7]. The engine distinguishes *forced*, *free*, *solver-undetermined*, and *inconsistent* as first-class outcomes.

5 Processes and structural amplitude constraints

Above the kernel, a process $i \to f$ receives a layered verdict from (a) exact gauge symmetries (charge, color), (b) the accidental global symmetries of the renormalizable theory (baryon and lepton number), (c) approximate symmetries (lepton flavor), (d) kinematics, and (e) a tree-level vertex-reachability search. The search makes "$\mu^- \to e^- \bar{\nu}_e \nu_\mu$ proceeds at tree level via W exchange" a *computed* statement rather than a conservation tautology, and correctly finds no tree diagram for $\mu^- \to e^- \gamma$.

Three further layers encode *amplitude-inspired* structural constraints: the quantum-number content of S-matrix consistency conditions, without introducing kinematics or evaluating amplitudes. The soft-theorem layer records the leading soft factors [8] and their Ward identities (a spin-1 emission is consistent iff the signed charge flow vanishes; a spin-2 emission iff the coupling is universal, reducing the Ward identity to momentum conservation). The factorization layer enumerates the internal lines through which a process factorizes (locality). The unitarity layer enumerates the multi-particle intermediate states a process can pass through (optical-theorem cuts), ordered by mass threshold. Each captures the structural content of an S-matrix consistency condition (charge conservation under soft emission, locality, the optical theorem) at the level of vertices and quantum numbers; none evaluates an amplitude, a residue, or a phase-space integral.

6 Candidate extensions and the full anomaly set

Within the flat-space, Lorentz-invariant, long-range mediator setting modeled here, a hypothesized new mediator is classified by the rank of the conserved current it couples to: a spin-s field couples to a rank-s current (a classification heuristic within the modeled class, not a theorem about all mediators), while the implemented coupling templates in this prototype are a scalar source (rank 0), a conserved gauge current (rank 1), and the stress-energy tensor (rank 2) [9, 10]. A massless mediator coupling universally to energy–momentum is therefore classified as spin 2. We treat massless higher-spin long-range couplings as outside the consistency class modeled here, rather than as targets for this structural search.

For a newly *gauged* $U(1)_X$ alongside hypercharge, the full anomaly set includes pure-X and *mixed* coefficients. Writing d_3, d_2 for the color and weak dimensions, T_c, T_w for the Dynkin indices, A_c for the $SU(3)$ cubic coefficient, and $s = \pm 1$ for chirality, the engine evaluates

$$
\begin{gathered}
[SU(3)^2 X] = \sum s\, d_2 T_c\, X, \qquad [SU(2)^2 X] = \sum s\, d_3 T_w\, X, \qquad [\mathrm{grav}^2 X] = \sum s\, d_3 d_2\, X, \\
[X^3] = \sum s\, d_3 d_2\, X^3, \qquad [Y^2 X] = \sum s\, d_3 d_2\, Y^2 X, \qquad [Y X^2] = \sum s\, d_3 d_2\, Y X^2, \qquad (2)
\end{gathered}
$$

together with the corresponding Y-sector and $[SU(3)^3]$ coefficients. The mixed terms (2) are load-bearing: a charge assignment can cancel every pure-X anomaly yet fail $[Y^2 X]$, so a single-$U(1)$ check can certify a still-anomalous model as consistent.

7 Minimal repair

Given a charge assignment that fails the conditions above, `nsm` searches for the smallest set of added fermions whose charges cancel the residual anomalies. The anomaly coefficients are linear, quadratic, and cubic in the unknown added charges, so the search is naturally posed for the SMT solver.

The menu is *tiered*. The simplest entry is the *sterile right-handed singlet* (color and weak singlet with $Y = 0$ and unknown X): neutral under everything except $U(1)_X$, such a field contributes only to $[\mathrm{grav}^2 X]$ and $[X^3]$ (each $\propto -X$ and $-X^3$ respectively), so cancelling a residual $(g, c) = ([\mathrm{grav}^2 X], [X^3])$ requires

$$
\sum_i X_i = g, \qquad \sum_i X_i^3 = c, \qquad (3)
$$

solved for gauged $B - L$ by a single ν_R. Beyond the singlet, the menu adds *SM-vector-like,* $U(1)_X$-*chiral* exotic pairs E, L, D, Q in definite SM representations with left/right charges $X_L \neq X_R$. Being vector-like under the Standard Model, each pair cancels its own SM anomalies; its chiral $U(1)_X$ split feeds the X-sector, with contributions $\propto (X_L - X_R)$ and $(X_L^3 - X_R^3)$. A hypercharged pair (E or L) thereby reaches the mixed $[Y^2 X]$, $[Y X^2]$, and a colored pair (D or Q) reaches $[SU(3)^2 X]$: precisely the anomalies a sterile singlet cannot touch. The chirality condition $X_L \neq X_R$ is essential; an X-vector-like pair would be inert.

coefficient	value	coefficient	value
$[SU(3)^3]$	0	$[\text{grav}^2 X]$	-1 **(fails)**
$[SU(3)^2 Y]$, $[SU(3)^2 X]$	0	$[Y^3]$	0
$[SU(2)^2 Y]$, $[SU(2)^2 X]$	0	$[Y^2 X]$, $[Y X^2]$	0
$[\text{grav}^2 Y]$	0	$[X^3]$	-1 **(fails)**

Table 2: The eleven anomaly coefficients for gauged $B-L$ with Standard-Model fermions only, as computed by `nsm`. Only the gravitational and cubic X-anomalies fail.

The engine returns the minimum *tier* that repairs the model (sterile singlet, then colorless exotic, then colored exotic), at minimal field count and then representation complexity. If a nonzero anomaly lies outside the reach of every menu item, the model is reported *structurally blocked* with that coefficient named; if every nonzero anomaly is reachable but no repair fits the field budget, it is reported *budget-limited* instead. The tool thus never claims a fix it cannot deliver, nor mistakes a budget limit for an obstruction.

The search emits a *solver-backed minimality record* rather than merely asserting minimality: for every field count below the one returned, the solver's verdict is recorded. An `unsat` verdict is a decision over the reals (no charges of that size satisfy (3), rational or otherwise), so a ladder of `unsat` rungs certifies that no smaller repair exists within the menu. A size the solver cannot decide (`unknown`), or one for which the solver returned a nonrational (algebraic) model (which does not prove that no rational solution exists), is recorded honestly and minimality is *not* claimed. This record's soundness rests only on that of the solver's nonlinear-real-arithmetic decision procedure (no false `unsat`), so "minimal within the menu" becomes a recorded result rather than an assertion. (We record the solver's verdicts, but not tracked unsat cores, SMT-LIB replay artifacts, or proof logs. An unsat core would identify a conflicting subset of assumptions; a proof log, where supported, would provide the stronger proof artifact.) For gauged $B-L$ it is immediate: the only smaller content is the unrepaired, inconsistent model, so a single ν_R is minimal.

The reported repair is minimal *within the menu and the field-count cost*; it is not unique, not a naturalness statement, and not a claim that the field exists.

8 Case study: gauging $B-L$

Assign each fermion its $B-L$ charge ($+1/3$ for quarks, -1 for leptons) and gauge it. The full anomaly set (Table 2) shows the model is inconsistent, and *why*: only $[\text{grav}^2 X]$ and $[X^3]$ fail, both equal to -1; every mixed and non-abelian coefficient already cancels.

By (3) with $g = c = -1$, the unique single-field repair is $X = -1$: one sterile right-handed singlet of $B-L$ charge -1, i.e. a right-handed neutrino. Re-running the full anomaly set on the repaired content returns all zeros. The engine has rediscovered the standard motivation for ν_R in gauged $B-L$ [11], without being told the answer. With ν_R in place the $B-L$ Z' is consistent, and the exposure layer reports it as resonantly produced in $e^+e^- \to Z'$, as an s-channel resonance/interference contribution to $e^+e^- \to \mu^+\mu^-$, and, being flavor-diagonal, as *not* a source of $\mu \to eZ'$. This single run threads gauged $B-L$, the full anomaly check, the minimal sterile repair, the ν_R motivation, and Z' phenomenology into one explained chain.

Gauged $B-L$ is only the sterile corner of the repair map. The same engine classifies the *minimum tier* of new matter required for any $U(1)_X$ charge assignment (Table 3), and a second worked case makes the tiering concrete. Assign $X = 1$ to e_R alone; the engine returns the anomaly

charge assignment	minimum repair tier	example repair
$B-L$	sterile-repairable	ν_R with $X=-1$
X on e_R only	colorless-exotic-repairable	an E_L/E_R chiral split
X on Q_L only	colored-exotic-required	a Q_L/Q_R chiral split

Table 3: The minimum repair tier for three $U(1)_X$ charge assignments, as classified by `nsm`. $B-L$ is the familiar sterile case; the others demand hypercharged or colored exotics that the sterile menu cannot supply.

vector

$$\big([\mathrm{grav}^2X],\,[Y^2X],\,[YX^2],\,[X^3]\big) = (-1,\,-4,\,+2,\,-1), \qquad [SU(3)^2X] = [SU(2)^2X] = 0.$$

A sterile singlet has $Y=0$, so it feeds only $[\mathrm{grav}^2X]$ and $[X^3]$ and *cannot* touch the mixed $[Y^2X]$ or $[YX^2]$; the engine reports the sterile menu as structurally blocked on exactly those two coefficients. The search is therefore forced up one tier, and the colorless menu repairs the model with one hypercharged E_L/E_R pair ($Y=-2$): two Weyl fields carrying $X_L=1$, $X_R=0$ in the signed left/right convention of Eq. (2), vector-like under the Standard Model and chiral under $U(1)_X$. With that pair included, all eleven coefficients re-verify to zero.

This example isolates anomaly repair. Invariance of the ordinary electron Yukawa interaction under the added $U(1)_X$ is a separate model-building constraint unless the scalar charges or scalar sector are included in the skeleton.

The minimality record is a genuine ladder: the one-field ($1N$) and two-field ($2N$) sterile attempts are both `unsat` (no lone singlet, and no pair of singlets, can generate the $[Y^2X]$ it must cancel), so the two-field colorless pair is certified minimal within the declared menu and cost function. A charge on Q_L alone instead breaks the colored $[SU(3)^2X]$, which only D or Q can reach, pushing the same machinery one tier further.

This does not discover particles; it systematically classifies the smallest completion within a declared representation menu and cost function. That turns the layer from a checker that validates one known result into an explorer that maps a space of anomalous extensions into minimal-completion classes, reporting what *kind* of new matter a broken gauge extension minimally demands.

A repair atlas. Because the classifier is mechanical, we can run it over a family of extensions rather than a single example. We enumerate every integer $U(1)_X$ charge assignment to one generation with $|q|\le 2$ and reduce rescaling/sign duplicates to primitive representatives (the tier is invariant under $X\to\lambda X$; it is also invariant under hypercharge mixing $X\to X+\alpha Y$, which we do not quotient but use as an internal consistency check), then classify each by its cheapest repair tier. The resulting census (Table 4) is a reproducible, basis-dependent slice: of the 1442 assignments distinct under sign and overall integer rescaling (one of which is the trivial zero generator, the lone "already-consistent" cell and not itself a gauged extension), 0.1% are sterile-repairable, 9.6% need only colorless exotics, 31.5% require colored exotics, and the remaining 58.7% are not repaired within a six-field budget.

Three caveats are built into the table. First, the budget-limited majority is a *field-budget artifact*, not an unrepairability theorem: with the full menu every failing anomaly is individually reachable (a necessary, not sufficient, condition for a joint repair), no cell is structurally blocked, and the budget-limited fraction falls monotonically as the budget grows (it has not plateaued at six fields). We therefore report budget-limited as "not found within the budget," not as a proof that a repair exists at larger size. Second, the percentages count a bounded integer box in one fixed (Y,X)

minimum repair tier	count	share
already-consistent	1	0.1%
sterile-repairable	1	0.1%
colorless-exotic-repairable	139	9.6%
colored-exotic-required	454	31.5%
budget-limited (> 6 fields)	847	58.7%

Table 4: Repair-atlas census of all 1442 sign- and rescaling-distinct (primitive, sign-fixed) integer $U(1)_X$ assignments to one generation with $|q| \leq 2$ in the fixed (Y, X) basis, each classified by the cheapest repair tier within a budget of six added fields (menu N/E/L/D/Q, solver timeout 8 s). The raw grid has 3125 cells (2.17× deduplication); 0 cells are structurally blocked and 81 (5.6%) carry an uncertified solver rung.

basis; they are not a measure over all $U(1)_X$ extensions, and a different choice of generator would re-slice the population. Third, 5.6% of cells carry an uncertified solver rung (an `unknown` or `sat-nonrational-model` verdict in the minimality record), which we report rather than suppress. With those stated, the atlas is the method claim made concrete: not one rediscovery, but a reproducible map of which kind of new matter a space of anomalous gauge extensions minimally demands.

9 Implementation

`nsm` is implemented in Python over the Z3 SMT solver as focused modules sharing one engine; the entire narrated pipeline runs via `python -m nsm`. Consistency conditions are compiled to solver assertions, and the routine that *checks* a concrete spectrum is the same one that *solves* for an unknown one: the anomaly-coefficient arithmetic is polymorphic in its charge arguments, returning exact rationals when fed rationals (a check) and symbolic expressions when fed solver variables (a constraint). Listing 1 shows a representative query.

Listing 1: Public API: forced hypercharges, and a minimal repair.

```
from fractions import Fraction as F
from nsm import derive_hypercharges, minimal_repair
from nsm.sm import standard_model_skeleton

r = derive_hypercharges(standard_model_skeleton())
r.hypercharges["Q_L"], r.is_unique     # -> (Fraction(1, 3), True)

# gauged B-L (charges as exact Fractions); repair searches for a fix
B_minus_L = (
    ("Q_L", F(1, 3)), ("u_R", F(1, 3)), ("d_R", F(1, 3)),
    ("L_L", F(-1)), ("e_R", F(-1)),
)
minimal_repair(B_minus_L).added         # -> [('nu_R', Fraction(-1, 1))]
```

The repair-atlas layer uses the same primitive over an integer charge grid: it reduces each vector to a primitive sign-fixed representative, keeps the raw and deduplicated denominators, classifies the representative by its minimum repair tier, and exports CSV, Markdown, or LaTeX summaries. Hypercharge-basis shifts are treated as an equivalence check rather than as a counting quotient, since they shear the integer lattice.

The kernel poses the hypercharge problem as a conjunction of assertions (each anomaly coefficient equal to zero, gauge invariance of each Yukawa coupling, and one normalization) and reads the unique assignment from the model. Whether that assignment *is* unique is decided by a second solve: assert the negation of the found solution and check satisfiability. `unsat` certifies uniqueness; `sat` exhibits a distinct second solution (such as the $u_R \leftrightarrow d_R$ ambiguity of Section 4); and `unknown` (the solver exceeding its budget on the nonlinear real arithmetic) is reported as a genuine third outcome rather than silently collapsed to false certainty. The same reachability and satisfiability idioms serve the dynamics, factorization, and unitarity layers. The minimal-repair search (Listing 2) encodes anomaly cancellation for each candidate menu combination, increasing the field count until the smallest solution is found and skipping nonrational solver models rather than failing on them.

Listing 2: Tiered repair: search the declared menu, smallest cost first.

```
for fields, complexity, combo in get_combinations(max_added, menu):
    if combo_cannot_reach_a_nonzero_anomaly(combo):
        record(combo, "unsat")          # sound prune, no solver call
        continue
    status, charges = solve_anomaly_equations(combo)
    record(combo, status)
    if status == "sat":
        repair = make_added_fields(combo, charges)
        assert verify_all_anomalies_vanish(repair)
        return repair                  # first sat is minimal in this menu/cost
```

10 Validation

Because the engine's value is its correctness, each layer was checked two ways. First, a test suite (171 tests, distributed with the code) pins the physics against hand-computed values, including *nonzero* anomaly coefficients, so that a wrong Dynkin or multiplicity factor which still cancels for the symmetric Standard Model cannot hide. Second, as internal quality assurance, each layer's results were independently re-derived and differential-tested against the implementation over randomized inputs; this process found and fixed real defects (for example, a non-terminating uniqueness search and a crash on nonrational solver models) and indicated that the full two-$U(1)$ check agrees with a single-$U(1)$ check wherever the latter applies while additionally catching mixed-anomaly failures, with no disagreement observed over 2×10^4 random charge assignments. This internal cross-checking is evidence of correctness, not a substitute for independent external reproduction; the test suite is the reproducible artifact.

Guaranteed versus observed. The engine rests on two trust anchors: anomaly coefficients for concrete spectra are computed in exact rational arithmetic, and the nontrivial solver inference is delegated to Z3's nonlinear-real-arithmetic decision procedure. A reported zero or nonzero coefficient is therefore exact, and an `unsat` verdict rules out a repair of that size over the reals. A ladder of all-`unsat` rungs is consequently a solver-backed guarantee of minimality within the declared menu and cost function. Everything else is reported as evidence, not proof: an `unknown` or `sat-nonrational-model` rung does not certify minimality, and cross-layer differential agreement is a test result rather than a theorem. For the anomaly and repair layers the tool either returns a solver-backed guarantee or labels the result uncertified; the structural layers (process, soft-factor, factorization, unitarity) are heuristic within their modeled class.

11 Limitations

`nsm` is a structural / quantum-number engine. It computes no cross sections, decay rates, or branching ratios; it has no kinematics (no momenta, spinors, or phase-space integrals; mass enters only as a crude threshold); it evaluates no amplitude values or pole residues; it performs no detector simulation and makes no statement of statistical significance or discovery. It does not derive masses, mixing angles, coupling constants, or the number of generations, which remain empirical inputs; an apparent "prediction" of one of these would be a bug. The vertex set is CKM-diagonal, composite hadrons are not expanded into constituents, and candidate gauge couplings are taken family-universal and vector-like. Several layer outputs are deliberately narrower than their names suggest: "color" conservation means *color triality* (a necessary, not sufficient, color-singlet test); baryon and lepton number refer to the modeled renormalizable vertex set; "kinematics" means decay thresholds and an $n \to 1$ phase-space check, not general momentum-space kinematics; and the unitarity layer enumerates *candidate* cut topologies with nominal mass thresholds, not cuts known to be open at a given $\sqrt{s}$. The one-line summary: `nsm` establishes *structural and quantum-number consistency*; it does not compute, measure, or predict any physical observable.

12 Conclusion

`nsm` demonstrates that a useful fraction of gauge-theory reasoning (what is forced, what is free, what is forbidden, and how to repair an inconsistent extension) can be made automatic, explainable, and mechanically checked, with a clean boundary between "structurally consistent" and "actually computed." Natural extensions include emitting tracked unsat cores, SMT-LIB replay artifacts, or solver proof logs for the minimality records, and, as a distinct phase that would cross the structural boundary, a spinor-helicity layer enabling genuine residue factorization and amplitude values.

Availability. The implementation, test suite, and a worked case study are in the accompanying repository.

References

[1] L. de Moura and N. Bjørner, *Z3: An Efficient SMT Solver*, in TACAS 2008, LNCS **4963** (Springer), pp. 337–340 (2008).

[2] A. Alloul, N. D. Christensen, C. Degrande, C. Duhr, and B. Fuks, *FeynRules 2.0–A complete toolbox for tree-level phenomenology*, Comput. Phys. Commun. **185**, 2250–2300 (2014).

[3] F. Staub, *SARAH 4: A tool for (not only SUSY) model builders*, Comput. Phys. Commun. **185**, 1773–1790 (2014).

[4] F. Lyonnet, I. Schienbein, F. Staub, and A. Wingerter, *PyR@TE: Renormalization Group Equations for General Gauge Theories*, Comput. Phys. Commun. **185**, 1130–1152 (2014).

[5] S. Kraml, S. Kulkarni, U. Laa, A. Lessa, W. Magerl, D. Proschofsky-Spindler, and W. Waltenberger, *SModelS: a tool for interpreting simplified-model results from the LHC and its application to supersymmetry*, Eur. Phys. J. C **74**, 2868 (2014).

[6] D. B. Costa, B. A. Dobrescu, and P. J. Fox, *General Solution to the $U(1)$ Anomaly Equations*, Phys. Rev. Lett. **123**, 151601 (2019).

[7] E. Witten, *An $SU(2)$ Anomaly*, Phys. Lett. B **117**, 324 (1982).

[8] S. Weinberg, *Infrared Photons and Gravitons*, Phys. Rev. **140**, B516 (1965).

[9] S. Coleman and J. Mandula, *All Possible Symmetries of the S Matrix*, Phys. Rev. **159**, 1251 (1967).

[10] S. Weinberg and E. Witten, *Limits on Massless Particles*, Phys. Lett. B **96**, 59 (1980).

[11] R. N. Mohapatra and R. E. Marshak, *Local $B-L$ Symmetry of Electroweak Interactions, Majorana Neutrinos and Neutron Oscillations*, Phys. Rev. Lett. **44**, 1316 (1980).